#  Anupam Raj

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## Education

**Ph.D. in Science Education (Pursuing)** 2025 – Present
*Teachers College, Columbia University, NY, USA*
*Dissertation Focus:* Understanding the Impact of AI on Teaching, Learning, and the Labor Market

**M.S. in STEM Education** 2023 – 2025
*University of Massachusetts Dartmouth, MA, USA*
*Capstone Project:* A Comparative Policy Analysis of India's National Education Policy 2020 and Contemporary U.S. Education Reforms

**M.S. in Teaching** 2019 – 2022
*University of Maine, Orono, ME, USA*
*Thesis:* Investigating the Teaching and Assessment Experiences of Maine Secondary Science Teachers During the COVID-19 Lockdown

**M.S. Physics** 2016 – 2018
*National Institute of Technology, Srinagar, J&K, India*
*Thesis:* Next Generation Se-alloyed CdTe Solar Cells at the University of Illinois, Chicago, USA

**B.S. Physics (Hons.)** 2013 – 2016
*Amity University, Noida, Uttar Pradesh, India*

## Current Projects

1. **Exploring the Experience and Perception of Black and Latino Youth in NYC with AI.** Spencer Foundation-funded project, Edmund W. Gordon Institute for Advanced Study, Teachers College, Columbia University. Serving as Lead Analyst, with PI Dr. Ezekiel Dixon-Román and Co-PI Dr. Nicole Sansone Ruiz.
2. **Epistemic Injustice in Teacher Perceptions of AI Research.** A mixed methods comparative review of how research on teachers' perceptions of AI differs across WEIRD and non-WEIRD countries. AEQUITAS Lab, Teachers College, Columbia University, with Dr. Renzhe Yu.
3. **The Student's Dilemma: Using Game Theory and Mechanism Design to Realign AI Use in Education.** A game-theoretic model of AI use in the classroom, with Dr. Andrew Gelman (Columbia) and Dr. Joseph McIntyre (Harvard); manuscript in preparation for *PNAS*.

## Publications

1. Raj, A., Miglani, A., Xu, T., Bosley-Smith, C., & McIntyre, J. (2026). Opportunity to grow, incentive to pursue: Future of U.S. vocational-technical jobs in the age of AI. *White Paper*. Pioneer Institute for Public Policy Research. (in press).
2. Raj, A., Miglani, A., Chawla, D., Bosley-Smith, C., & McIntyre, J. (2026). Fewer students, greater demand: What Massachusetts can learn from its vocational technical schools. *White Paper*. Pioneer Institute for Public Policy Research.
3. Raj, A. (2026). Reimagining learning as a 3-dimensional vector: Integrating behaviorism, cognitivism, and social constructivism. In *Proceedings of the Annual Meeting of the International Society of the Learning Sciences (ISLS) 2026*. International Society of the Learning Sciences. (in press).
4. Raj, A., & Wittmann, M. C. (2026). Teacher perception and practices of the Next Generation Science Standards (NGSS): Insights from a New England study. In *Proceedings of the Annual Meeting of the*

*International Society of the Learning Sciences (ISLS) 2026*. International Society of the Learning Sciences. (in press).

5. Raj, A., Bains, S., & Kaur, A. (2026). Purpose and paradox in India's National Education Policy 2020: The catch-22s of learning reform. In *Proceedings of the Annual Meeting of the International Society of the Learning Sciences (ISLS) 2026*. International Society of the Learning Sciences. (in press).
6. Raj, A., Stroup, W., & Kayumova, S. (2025). Stories, printing press, internet, and now ChatGPT: Examined via the new SMART framework. In *Proceedings of the 19th International Conference on Computer-Supported Collaborative Learning – CSCL 2025* (pp. 445–449). International Society of the Learning Sciences.
7. Raj, A., Kushimo, K., & McIntyre, J. C. (2025). Educational attainment, diversity, and trust dynamics in Nigeria. In *Proceedings of the 19th International Conference of the Learning Sciences – ISLS 2025* (pp. 1115–1122). International Society of the Learning Sciences.
8. Raj, A., & Kayumova, S. (2025). Zone of playful development: Play for transformational equity in multilingual STEM classrooms. In *Proceedings of the 19th International Conference of the Learning Sciences – ISLS 2025* (pp. 2105–2109). International Society of the Learning Sciences.
9. Raj, A., Wittmann, M., & Kayumova, S. (2025). Resilient and transformative journey of teachers through the pandemic and beyond. In *Proceedings of the 19th International Conference of the Learning Sciences – ISLS 2025* (pp. 2115–2119). International Society of the Learning Sciences.
10. Raj, A., Wittmann, M., & Kayumova, S. (2025). Rethinking differentiated instruction: Lessons from the pandemic for today's diverse classrooms. In *Proceedings of the 19th International Conference of the Learning Sciences – ISLS 2025* (pp. 3200–3203). International Society of the Learning Sciences.
11. Raj, A., Ashique, S. M. R., & Bosley-Smith, C. (2024). Lingering effects of the pandemic on student performance in Massachusetts, USA. *The International Journal of Engineering and Science*, 13(6), 38–41.
12. Raj, A., Wittmann, M., & Kayumova, S. (2024). Being fair & equitable: A qualitative study of science teachers' shifts in priorities and expectations during COVID-19. In *Proceedings of the 18th International Conference of the Learning Sciences – ICLS 2024* (pp. 2283–2284). International Society of the Learning Sciences.
13. Kayumova, S., Raj, A., & Harper, A. (2024). I see myself as a Science Person: Insights into science identity development among emergent multilingual youth. In *Proceedings of the 18th International Conference of the Learning Sciences – ICLS 2024* (pp. 2507–2508). International Society of the Learning Sciences.
14. Raj, A. (2022). *Investigating the Teaching and Assessment Experiences of Maine Secondary Science Teachers During the COVID-19 Lockdown*. The University of Maine.

## Conference Presentations

**ISLS Annual Meeting 2026** — Irvine, CA, USA
Presented three papers, in press (see Publications).

**ISLS Annual Meeting 2025 (incl. ICLS & CSCL)** — Helsinki, Finland
Presented five papers (see Publications).

**ICLS 2024** — Buffalo, NY, USA
Presented two papers at the 18th International Conference of the Learning Sciences (see Publications).

**NARST 2025 Annual International Conference** — National Harbor, MD, USA
"Supporting Emergent Bilingual Students' Understanding of Energy Through Equitable Teaching Practices." Accepted as a full paper presentation.

## PROFESSIONAL SERVICE

**Reviewer, ISLS Annual Meeting** 2026
Reviewed three papers.

**Reviewer, ISLS Annual Meeting** 2025
Reviewed three papers.

## TEACHING EXPERIENCES

**Teaching Assistant** Summer 2026
*Graduate School of Arts and Sciences (GSAS), Columbia University, 535 W 116th St., New York, USA*
Supporting *POLS UN3720: Scope and Methods* (RCTs, and Quasi-Experimental designs including Natural Experiments, Instrumental Variables, Difference-in-Differences, and Regression Discontinuity) with Dr. Christina Chiopris. Role includes grading, student support, and discussion facilitation.

**Teaching Assistant** Spring 2026
*Graduate School of Arts and Sciences (GSAS), Columbia University, 535 W 116th St., New York, USA*
Supporting *Rationalizing the World: American Social Science* with Professor Andrew Gelman. Role includes grading, student support, and discussion facilitation.

**Teaching Assistant** Fall 2025
*Graduate School of Arts and Sciences (GSAS), Columbia University, 535 W 116th St., New York, USA*
Supporting *Introduction to Quantitative Analysis I* for Political Science with Professor Robert Shapiro. Role includes grading, student support, and discussion facilitation.

**Physics Tutor** Spring 2025
*946 Washington St, Dorchester Center, MA, USA*
Taught physics to high school students on weekends. This volunteer role was unpaid and undertaken to promote equity and access in STEM fields by contributing to the learning of marginalized students.

**Turnaround Physics Teaching** Fall 2022 – Summer 2023
*Excel High School, 95 G St., Boston, MA, USA*
Full-time classroom teacher for 9th grade physics and Instructional Team Leader for the Science Department.

**Student-Teaching** Spring 2022
*Portland High School, 284 Cumberland Ave, Portland, ME, USA*
Taught physics and engineering to high school students during a 15 week-long practicum, with collaboration and active mentor-teacher feedback.

**Teaching Assistant** Fall 2021
*Physics Dept, University of Maine, Orono, ME, USA*
Taught Intro to Physics course as a Lab and Recitation Instructor. Job roles included teaching, grading, proctoring, and mentoring.

**Summer School Instructor** Summer 2021
*Leonard Middle School, 156 Oak St., Old Town, ME, USA*
Part of the team that planned and executed the summer school program. Job roles included teaching English, STEM, Mathematics, Yoga, and Dancing.

**Teaching Assistant** 2019 – 2020
*Math Dept, University of Maine, Orono, ME, USA*
Taught pre-calculus as a TA for 2 semesters (Fall 2019 and Spring 2020) and Calculus 1 (Fall 2020). Responsibilities included assisting the lecturer in teaching, grading, proctoring, and mentoring.

**CITL Bootcamp for Graduate Teaching Assistants** Spring 2020

*University of Maine, Orono, ME, USA*
Participated in a weekly workshop geared toward improving teaching skills using the latest strategies, research, and tools, with the Center for Innovation in Teaching and Learning (CITL).

**4-H STEM Ambassador** Spring 2020
*University of Maine Cooperative Extension, Orono, ME, USA*
Facilitated weekly experiential learning workshops for middle school students at Hampden Middle School in Maine. The program was terminated due to COVID-19.

**Physics Faculty** April – Sept. 2017
*Aakash Institute, Old Zero Bridge, Rajbagh, Srinagar, J&K 190008, India*
Engaged and inspired secondary school students to read and explore Physics. Part-time.

## Research Experiences

**Interim Research Support (Research Assistant)** 2025 – 2026
*Edmund W. Gordon Institute for Advanced Study, Teachers College, Columbia University*
Working with Dr. Ezekiel Dixon-Román and Dr. Nicole Sansone Ruiz on a Spencer Foundation-funded study of Black and Latino youth's experience and perception of AI in New York City.

**Research Assistant** Fall 2025 – Present
*AEQUITAS Lab, Teachers College, Columbia University, 525 W 120th St, NY, USA*
Working with Dr. Renzhe Yu on projects related to AI, ethics, and education, with focus on teacher perceptions and algorithmic bias.

**Research Consultant** 2025 – Present
*Pioneer Institute, 185 Devonshire St, Boston, MA, USA*
Co-authored working papers to understand the success of vocational-technical schools in Massachusetts and the future of vocational-technical jobs in the age of AI.

**Research Collaborator** Fall 2024 – Present
*Harvard Graduate School of Education, 13 Appian Way, Cambridge, MA, USA*
Started as an intern auditing an Intermediate Linear Regression class while working on multiple research projects with Dr. Joseph McIntyre involving DESE, PISA, and Afrobarometer datasets. The research collaboration is ongoing.

**Research Assistant** Fall 2021
*RiSE Center (Research in STEM Edu.), University of Maine, Orono, ME, USA*
Worked as an RA for the INSPIRES project. Job roles included literature review, transcribing, and coding.

**Research Intern in Physics Education Research Lab** Jan – Feb 2019
*Homi Bhabha Center for Science Education, TIFR, India*
Developed low-cost physics lab demonstrations and activity sheets from everyday materials for under-privileged schools, while working as a short-term visitor.

## Leadership Experiences

**Department Senator** 2025 – 2026
*Student Senate, Teachers College, Columbia University, 525 W 120th St, NY, USA*
Elected representative for the Department of Mathematics, Science, and Technology; engaged in policy discussions, advocacy, and resource allocation for PhD student needs.

**Science Instructional Leadership Team Leader** Fall 2023 – Spring 2024
*Excel High School, 95 G St., Boston, MA, USA*
Facilitated Core Planning Team (CPT) meetings for the science department.

**South Asian Association of Maine (SAAM) Secretary** Fall 2021
*University of Maine, Orono, ME, USA*
Assisted SAAM in cultural events for UMaine students. Responsibilities included increasing the outreach and assisting in organizing events.

**RiSE Research Group Executive Board Member** Spring 2020
*RiSE, University of Maine, Orono, ME, USA*
Secretary of the Research in STEM Education group at the University of Maine. Responsibilities included setting up meetings and writing the agenda, sending reminders, and taking notes.

**Vice President – Incoming Global Volunteer** 2016
*AIESEC in Delhi University, H-4/13, Malviya Nagar, New Delhi 110017, India*
Coordinated operations, project management, and training. Facilitated the AIESEC International student exchange program by managing membership, projects, and international interns for overall leadership development.

**Intern for MS Orientation Program** Nov 2016
*Inter University Accelerator Centre (IUAC), New Delhi, India*
Intern for the Accelerator Physics and Spiral Bunchers project.

**Global Intern – AIESEC Global Volunteer Program** May 2015
*Project JUNKIRI, Kathmandu, Nepal*
Social project to help school students overcome the fear of attending school after the 2015 earthquake.

## Languages, Skills, and Hobbies

**Fluent** in English and Hindi (native). Conversational in Maithili, Bhojpuri, Bengali, Urdu, Haryanvi, & Punjabi. Currently learning Spanish.

**Basic Proficiency** with Python, R, SPSS, MS Office, HTML, LaTeX, C, and Java programming.

**Hobbies** include Dancing (WCS & Bachata), Sports (Tennis & Volleyball), Reading, and Meditation.

## Achievements

**Teachers College Zankel Fellowship** 2026-2027
Teachers College, Columbia University
*Fellowship to be completed with Dr. Oren Pizmony-Levy*

**Global Policy Fellow for Education in Emergencies** 2025-2026
Teachers College, Columbia University

**Science Education Endowment Scholarship** 2025-2026
Teachers College, Columbia University

**Emerging Leader Financial Award** 2025-2028
Teachers College, Columbia University

**Advocated for Change** 2023
in Boston Public Schools to hire and support international teachers
`https://www.dotnews.com/2023/new-program-aims-keep-teachers-visa-issues-city-classrooms`

**Outstanding Graduate Student Award** 2022
RiSE department, University of Maine

**Featured in INSPIRE Grad Student Spotlight** 2022
while working at the University of Maine
`https://umaine.edu/epscor/2022/08/22/inspires-graduate-student-spotlight-anupam-raj/`

**Advocated for International Students** 2020
in the USA during the COVID-19 lockdown ICE rule
https://www.mainepublic.org/nation/2020-07-12/foreign-students-in-maine-say-ice-rule-forces-them-to-risk-their-health-to-stay-in-the-country

**3-Year Assistantship** 2019
for MS in Teaching Program, University of Maine, USA

**Selected for the Teach for India** 2019
2019–2021 cohort, India

**Best National Portfolio Award** 2017
to iGV-, Daman and Diu, India (AIESEC in Delhi University at AIESEC NLS Conference)

**Qualified the IIT Joint Admission Test (JAM)** 2016
for M.Sc. Physics, India

**Silver Medalist** 2016
for Class of 2016 (Bachelor of Science), Amity University, Noida, India

**1st Position** 2014
Flick Click Photography Competition, Youth Fest, Amity University, Noida, India

**Commendation Certificate** 2012
for second runner up in Metallica (Science Quiz) at IIM, New Delhi, India

**Environment Captain** 2011-2013
Student Executive Body, High School (S.B.D.A.V)

**Financial Scholarship** 2011
(50% fee waiver), High School, for 10 CGPA in SSE

**Qualified the Aakash National Talent Hunt Exam** 2010

## References

**Dr. Felicia Mensah**
Professor, Teachers College, Columbia University, USA
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https://www.tc.columbia.edu/faculty/fm2140/

**Dr. Michael C. Wittmann**
Head of Education at the American Physical Society, USA
wittmann@aps.org
https://aaas-iuse.org/team/michael-wittmann/

**Dr. Joseph Clark McIntyre**
Lecturer, Harvard Graduate School of Education (HGSE), USA
joseph_mcintyre@gse.harvard.edu
https://www.gse.harvard.edu/directory/faculty/joseph-mcintyre

## Online Profiles

- LinkedIn: https://www.linkedin.com/in/anupam-raj-119124382
- Google Scholar: https://scholar.google.com/citations?user=cILRfwMAAAAJ&hl=en
- ResearchGate: https://www.researchgate.net/profile/Anupam-Raj